Ismail Kuru, Ph.D.

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Employment History

2026(Feb) –	**Postdoctoral Researcher.** Aarhus University, Department of Computer Science, Center for Basic Research in Program Verification. Working with Prof. Lars Birkedal and Prof. Jean Pichon-Pharabod.
2026(Oct) –	**Postdoctoral Researcher.** Center for R&D on Secure Computer Systems (CRADSEC), Research Organization of Information and Systems (ROIS), Tokyo, Japan. Working with Prof. Taro Sekiyama.
2026(Apr) – 2026(Sep)	**Visiting Researcher.** National Institute of Informatics, Tokyo, Japan. Hosted by Prof. Taro Sekiyama; co-advised a summer research intern.
2025(July) – 2026(Feb)	**Postdoctoral Researcher.** Drexel University, College of Computing & Informatics.
2016(Oct) – 2025(June)	**Doctoral Research Fellow.** Drexel University, College of Computing & Informatics.
2019(Apr) – 2019(June)	**Research Intern.** BedRock Systems.
2016(Jan) – 2016(Sep)	**Senior Software Engineer.** Crytek GmbH.
2014(Sep) – 2015(Feb)	**Research Assistant.** Koc University. Advisor: Dr. Serdar Tasiran.
2014(Apr) – 2014(Aug)	**Research Intern.** Microsoft Research. Advisors: Dr. Matthew J. Parkinson, Ben Hall and Serdar Tasiran.
2013(Sep) – 2014(Mar)	**Visiting Research Student.** University of Washington. Advisor: Dr. Dan Grossman.
2012(Sep) – 2013(Sep)	**Research Assistant.** Koc University.
2011(Mar) – 2011(Sep)	**Software Developer.** GNU Compiler Collection, Google Summer of Code 2011. Advisor: Dr. Albert Cohen.

Education

2018 – 2025 **Ph.D., Drexel University, U.S.A** in Software Reliability.
Thesis: *Modal Abstractions for Operating System Kernels*
Advisor: *Dr. Colin S. Gordon*
Funding: NSF Career Project (*link*)

2016 – 2018 **M.Sc. Computer Science, Drexel University, U.S.A.**
Internships: BedRock Systems
Coursework Completed

2015 **M.Sc. Computer Science, Koc University, Turkey** in Software Verification.
Thesis title: *Static Methods for Checking Correctness of Programs on Relaxed Memory Systems.*
Advisor: *Dr. Serdar Tasiran*
Scholarship: 1 of 18 Microsoft Research Europe PhD Scholarships (*link*)
Internships: University of Washington Visit and MSR Cambridge
Coursework Completed in Technical University of Munich, Germany.

Teaching and Mentoring

Drexel
- **Teaching Assistant**, CS 360 Programming Languages; led discussion sections.
- **Teaching Assistant**, SE 311 Software Engineering.

Koc
- **Teaching Assistant**, Software Engineering, Koc University.

2026
- **Research Advisor**, Keisuke Nishimura (research intern, National Institute of Informatics, summer 2026), program logic for inline assembly in Iris; co-advised with Taro Sekiyama through weekly meetings.

2021 –
- **Long-Term Mentor**, SIGPLAN-M (ACM SIGPLAN mentoring program), mentoring graduate and senior undergraduate students in programming languages research.

Skills

Languages	Strong reading, writing and speaking in English and Turkish. Intermediate German.
SMT Based Verification	VCC, Boogie, QED.
Model Checking	NuSMV.
Type Theory Based Verification	Rocq (formerly Coq), Iris.

Miscellaneous Experience

Awards and Achievements

2018
- **Scholarship**. DeepSpec Summer School Scholarship Princeton U.S.A, 2018.

2017
- **Scholarship**. PLMW Scholarship for ICFP'17 Oxford U.K, 2017.

2014
- **Travel Grant**. Microsoft Research Visitor Grants(Multiple) Hosted by Dr. Matthew Parkinson, 2014.

2011–2014
- **Graduate School Scholarship**. Received one of 18 Microsoft Research Europe PhD Scholarship awards, 2011-2014.

2012
- **Travel Grant**. from Koc University for VCLA Winter School Vienna, 2012.

2011
- **Travel Grant**. Inria-Paris Visitor Grants(Multiple) Hosted by Dr. Albert Cohen, 2011.

Summer/Winter Schools

2018
- DeepSpec Summer School Princeton U.S.A, 2018.

2017
- PLMW for ICFP'17 Oxford U.K, 2017.
- OPLSS Summer School Oregon U.S.A, 2017.

2012
- ACACES Eighth International Summer School Italy, 2012.
- Microsoft Research Ph.D. Summer School Cambridge U.K, 2012.
- Vienna Center for Logic and Algorithms Winter School on Verification Wien Austria, 2012.

2010
- ACACES Sixth International Summer School Barcelona Spain, 2010.

Invited Talks

2025
- *Principles and Practices for Foundational Operating System Kernel Verification*. RIOS Group, National Institute of Informatics, Tokyo, Japan, November 2025.

Service and Volunteering

2021
- SIGPLAN-M Long Term Mentor (link)

2020
- POPL'20 Artifact Evaluation Committee.

Miscellaneous Experience (continued)

2018 PLDI'18 Student Volunteer.

2013 SEFM'13 Subreviewer.

SAS'13 Subreviewer.

2012 RV'12 Subreviewer.

Manuscripts Under Submission

2026 Zongyuan Liu, Freja Marott Crawford, **Ismail Kuru**, Jean Pichon-Pharabod, and Lars Birkedal. *Modular and Tractable Verification of a Realistic Ticket Lock on Mixed-Size Arm in Separation Logic.* Under submission.

Ismail Kuru and Colin S. Gordon. *Resource Morphisms for Protocol Indistinguishability.* Under submission.

Ismail Kuru and Colin S. Gordon. *ModFs: Modal Abstractions for Snapshotting Copy-On-Write Filesystems.* Under submission.

Ismail Kuru and Colin S. Gordon. *Deferred Reclamation Is One Discipline: A Scheme-Agnostic Type System for Memory Safety, Mechanized.* Under submission.

Manuscripts In Preparation

2026 **Ismail Kuru** and Colin Stebbins Gordon. *specNL: Protocols, Resources, Contexts and Specification from Natural Language.* In preparation.

Keisuke Nishimura, **Ismail Kuru**, and Taro Sekiyama. *A Program Logic for Inline Assembly.* In preparation.

Ongoing Projects

2026 – **Verified trusted execution environments on RISC-V** (with Taro Sekiyama, CRADSEC/ROIS). An Iris instantiation over the Sail RISC-V semantics with an isolation-domain modality enforced by physical memory protection, targeting a Keystone-style security monitor and the GlobalPlatform TEE interface.

2025 – **iIris: a layered Iris program logic for x86-64**, with assembly, macro, and intermediate layers composing through an inline-assembly boundary, a verified encoder/decoder and compiler, and a port of the modal abstractions to a verified xv6 kernel on RISC-V. Nearing completion.

Research Publications

Conference Proceedings

1. Kuru, I., & Gordon, C. S. (2025a). Modal Abstractions for Virtualizing Memory Addresses. *OOPSLA 2025*.
2. Kuru, I., & Gordon, C. S. (2019). A Type System for Read-Copy-Update Concurrency [Acceptance Rate 32.6% (28/86).]. *European Symposium on Programming (ESOP 2019)*. https://doi.org/10.1007/978-3-030-17184-1_4
3. Kuru, I., Matar, H. S., Cristal, A., Kestor, G., & Unsal, O. (2013). Parv: Parallelizing runtime detection and prevention of concurrency errors. In S. Qadeer & S. Tasiran (Eds.), *Runtime verification* (pp. 42–47). Springer Berlin Heidelberg.

Thesis

1. Kuru, I. (2025). *Modal abstractions for operating system kernels* [Doctoral dissertation, Drexel University]. Drexel University. https://doi.org/10.17918/00010934

Books and Chapters

1. Cristal, A., Ozkan, B. K., Cohen, E., Kestor, G., Kuru, I., Unsal, O., Tasiran, S., Mutluergil, S. O., & Elmas, T. (2015). *Verification tools for transactional programs* (R. Guerraoui & P. Romano, Eds.). Springer International Publishing. https://doi.org/10.1007/978-3-319-14720-8_14

Workshop

1. Kuru, I., & Gordon, C. (13th Workshop on Programming Languages and Operating Systems PLOS 2025 – Co-located with SOSP 2025). *Modal verification patterns for systems software*.
2. Kuru, I., Kulahcioglu Ozkan, B., Mutluergil, S. O., Tasiran, S., Elmas, T., & Cohen, E. (Proceedings of the 9th ACM SIGPLAN Workshop on Transactional Computing, TRANSACT 2014). *Verifying programs under snapshot isolation and similar relaxed consistency models*.
3. Matar, H. S., Kuru, I., Tasiran, S., & Dementiev, R. (5th Workshop on Determinism and Correctness in Parallel Programming, WoDet 2014). *Accelerating precise race detection using commercially-available hardware transactional memory support*.

Tech Report

1. Kuru, I., & Gordon, C. S. (2025b). Modal verification patterns for systems. https://arxiv.org/abs/2506.01719
2. Kuru, I., & Gordon, C. S. (2024). Modal abstractions for virtualizing memory addresses. https://arxiv.org/abs/2307.14471
3. Kuru, I., & Gordon, C. S. (2018). Safe deferred memory reclamation with types. http://arxiv.org/abs/1811.11853

References

Available on Request